

Indirect FM → Direct PM Generation (Circuits)

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Disadvantage of Direct FM Generation

- Carrier generation is directly affected by modulating signal- not easy to get high order stability in carrier frequency

→ *change*

- Drift in the carrier frequency which needs to be stabilized

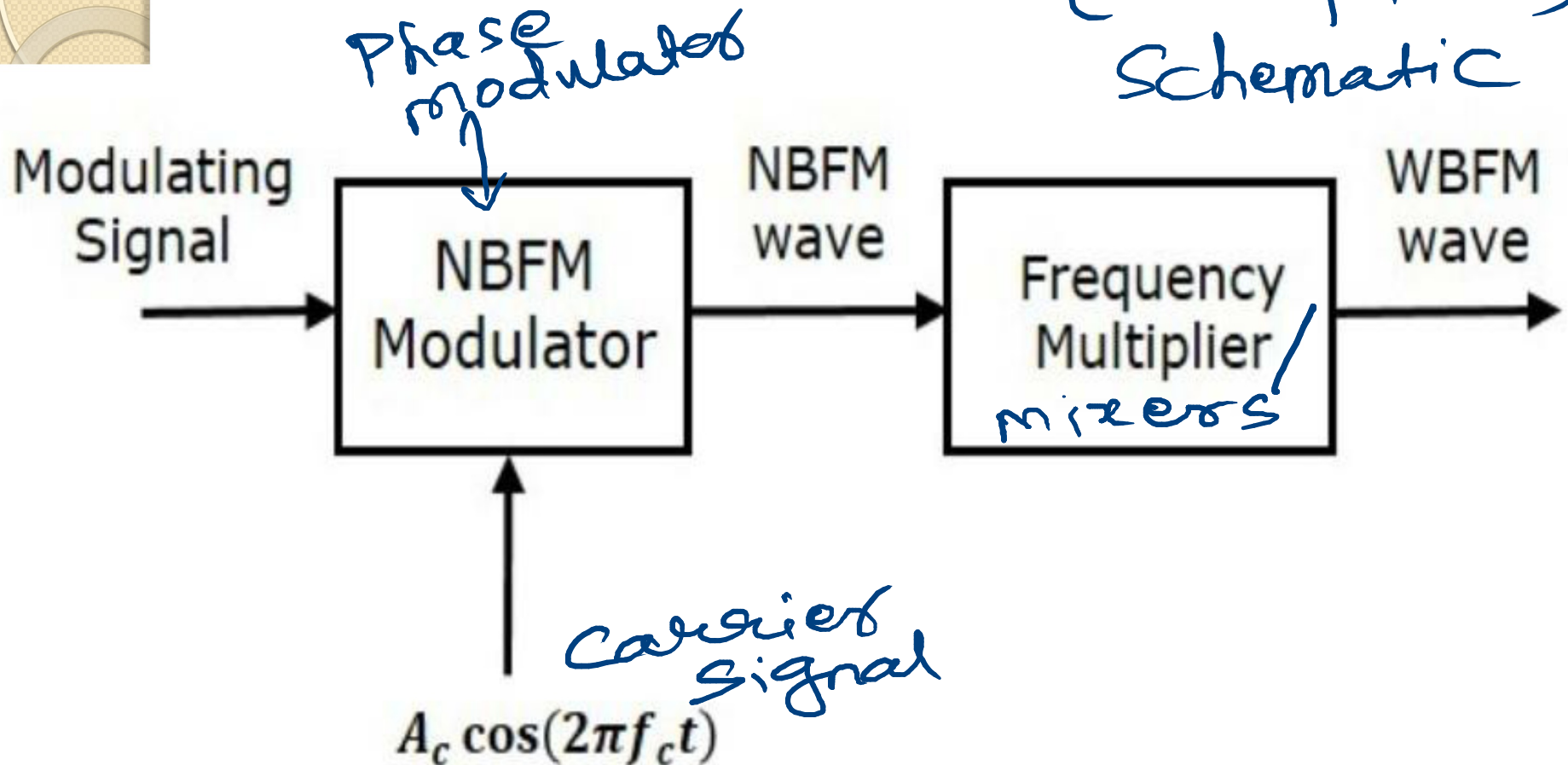
Indirect FM Generation

Armstrong Method- Divided in 3 parts

- Part 1: Generate NBFM signal using phase modulator (PM)
 → Narrow band FM
- Part 2: Implementation of Phase Modulator
- Part 3: Use of frequency multipliers, mixers and amplifiers
 → Convert NBFM signal into WBFM

Basic Block Diagram

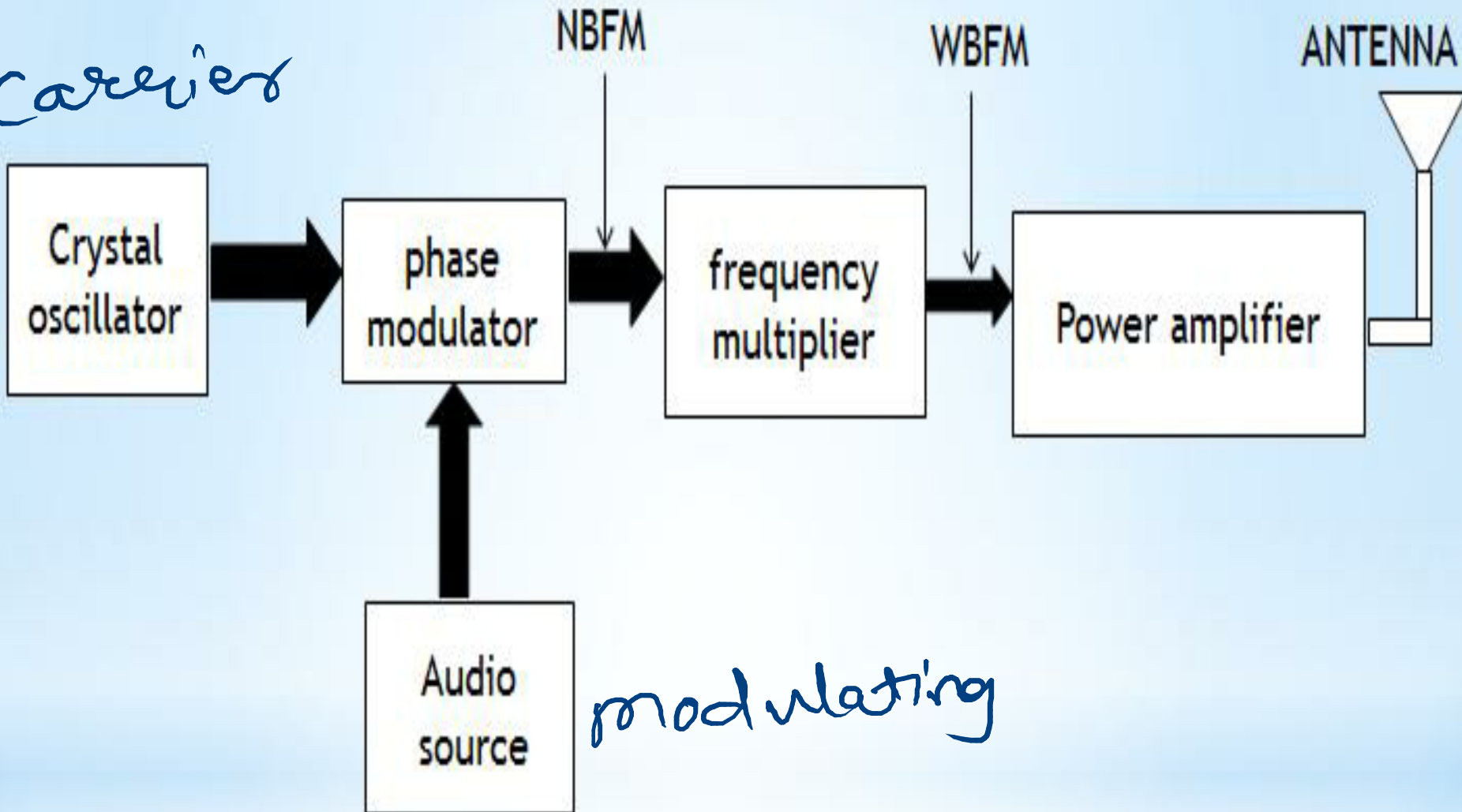
(Conceptual)
Schematic



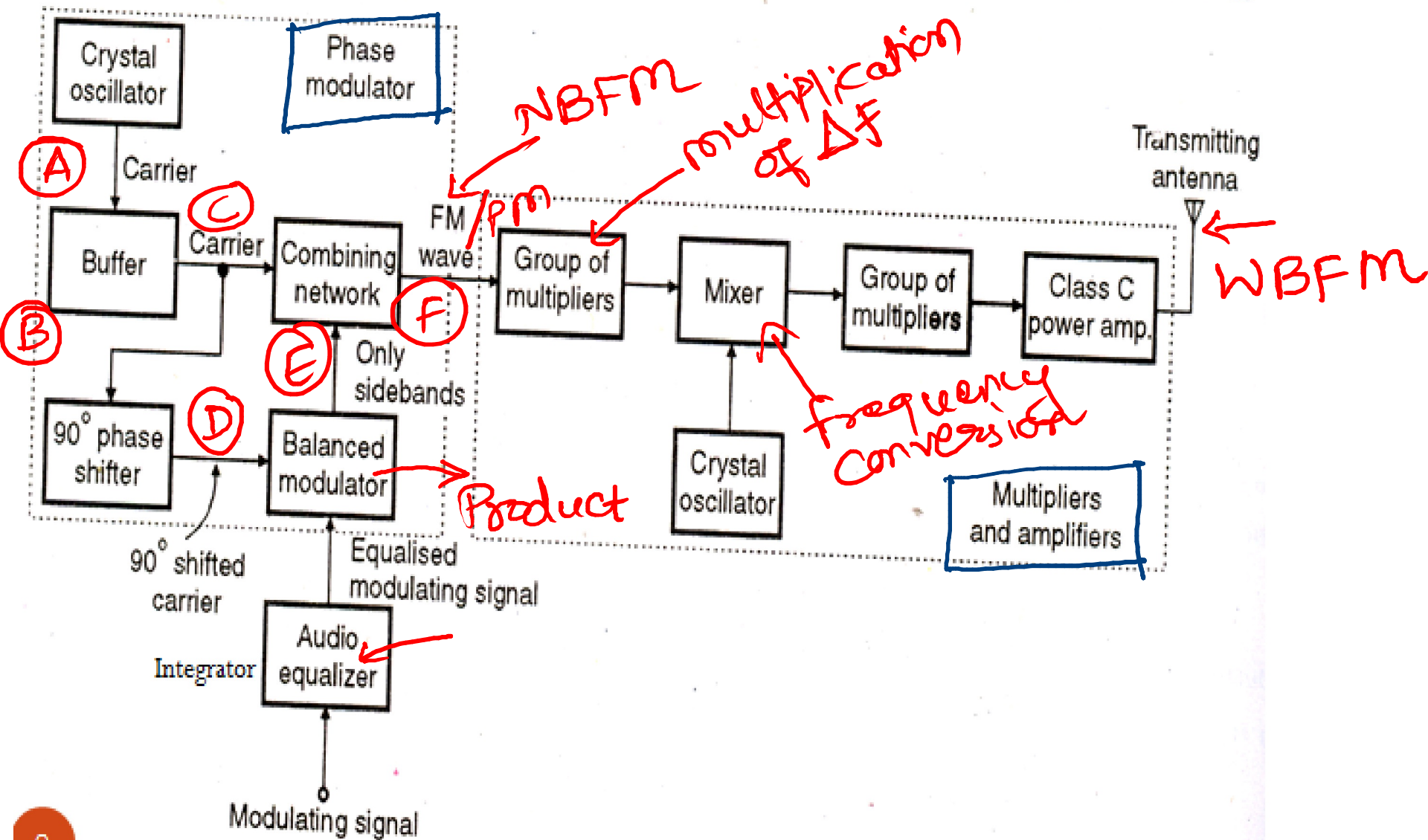
Basic Block Diagram

(Conceptual)

Carrier



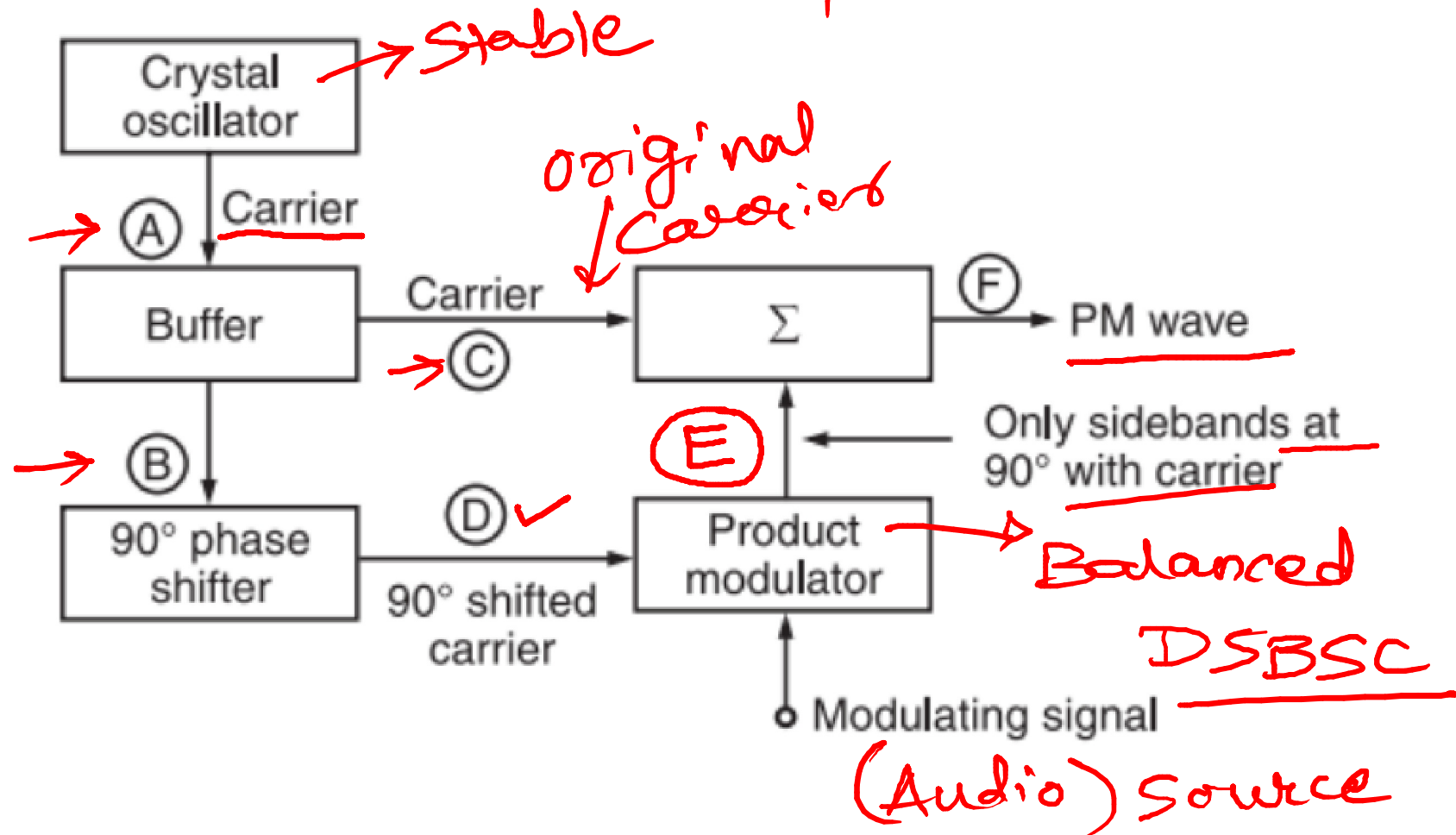
Indirect FM Generation



Indirect FM Generation

Phase Modulator

part - I & Part - II



Phasor Diagram- Generation of PM

output of crystal

Unmodulated carrier

At A, B, C

Unmodulated carrier

90° phase shifter output

At D

90° shifted carrier

Balanced modulator output

At E

USB

Suppressed carrier

LSB

Resultant of sidebands

At F

Combining network output

Resultant of sidebands

USB

LSB

Resultant at the output of combining network

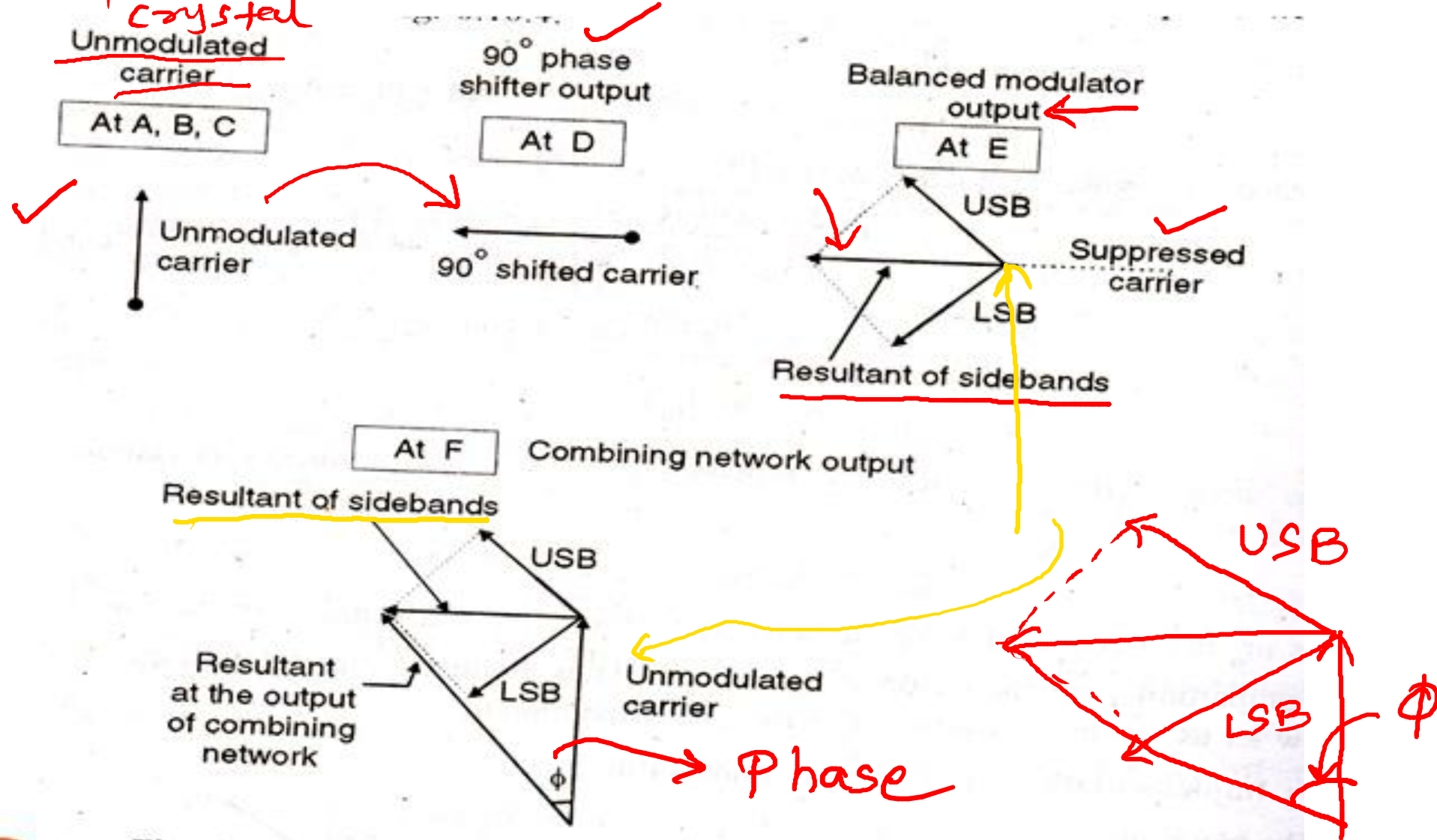
Unmodulated carrier

Phase

USB

LSB

ϕ

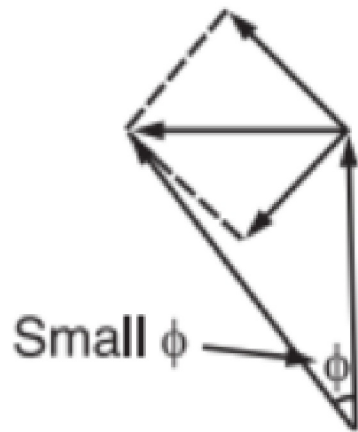


Phasor Diagram

Effect of Modulation Index on Frequency

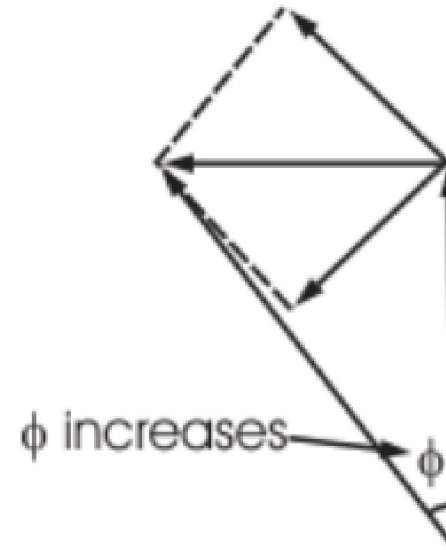
Low modulating
Signal

Low modulation index



High modulating
Signal

High modulation index





PM Modulators (Direct PM) Circuits

